

THATO BARUTI

tel: +44 07759 037559 Email: thato.baruti.619@cranfield.ac.uk
Linkedin: <https://www.linkedin.com/in/thato-baruti-63b290222/>

PERSONAL STATEMENT

A motivated and analytical Aerospace Engineering graduate and MSc Computational Fluid Dynamics candidate with strong capability in **modelling & simulation, digital tool development, and prototype led validation**. Experienced in translating specifications into high quality outputs (simulations, technical visuals, and reports) that enable **commercial decision making by technical and non-technical stakeholders**. Hands-on manufacturing and testing background (laser cutting, composites machining, 3D printing) with a quality focused mindset and the ability to learn quickly under tight time constraints. Royal Aeronautical Society (RAeS) Student Affiliate.

KEY ACHIEVEMENTS

- Awarded University of Sheffield International Undergraduate Scholarship and Botswana Government Top Achievers Scholarship for academic excellence.
- Achieved a **77% distinction** in Managing Engineering Projects and Teams, demonstrating strong project leadership capability.
- Designed and optimised a UAS fuselage structure, improving **stiffness to weight ratio by 35%** while dropping **material cost by 44%** through laser-cut composite fabrication.
- Increased drone flight stability by **25%** and extended battery endurance by **15%** leveraging CFD and FEA-validated design iterations.
- Ranked **Top 10 nationally** among 30+ UK university teams in the UKSEDS Rocketry Competition.

EDUCATION

MSc in Computational Fluid Dynamics, Cranfield University, Cranfield, UK **Oct 2025 – Dec 2026 (Expected)**

Selected modules: Turbulence Modelling, Numerical Methods & High Performance Computing (HPC), Grid Generation, Data Analysis & Uncertainty, Numerical Modelling

BEng Aerospace Engineering, University of Sheffield, Sheffield, UK **Jun 2025**

Selected study: Aerodynamic Design, Aircraft Dynamics & Control, Aerospace Design, Project Management, Accounting & Law for Engineers

Cambridge International A-Levels, John Mackenzie High School, Botswana **Jan 2020 – Nov 2021**

Mathematics (A), Physics (A), Chemistry (A), Biology (B)

IELTS Certificate | Overall Band Score: 7.5

MODELLING & SIMULATION HIGHLIGHTS

- **Simulation delivery mindset:** interpret specifications, define assumptions, configure models, monitor convergence, and validate outputs against expected behaviour or tests.
- **Digital assets:** generate clear technical visuals (plots, images, animations) using CFD post-processing tools; package outputs to specification for stakeholder use.
- **Verification/quality focus:** structured checks for consistency, accuracy, and reproducibility; documentation suitable for review/approval and handover.
- **Testing/validation orientation:** experience supporting prototype builds and performance assessment; comfortable linking model predictions to practical test outcomes.
- **Data & ML (DAPP):** implemented a **machine learning workflow in MATLAB** (data preparation, training/validation, metrics) to enhance prediction and decision-making under uncertainty.
- **Software skills:** coding in Python/MATLAB/FORTRAN/C++ for analysis automation, lightweight tools, and repeatable engineering workflows.

WORK EXPERIENCE

Event Steward **Jul 2023 – Jul 2023**
SIERRA 1 SECURITY STEWARDING, Silverstone, UK *Formula 1 Silverstone & Yorkshire Balloon Fiesta*

- Controlled crowd safety and flow at a major sporting event of over **140,000 attendees**, decreasing queue times by **20%**; received supervisor commendation for composure and leadership under pressure.
- Developed effective communication strategies across varied audience demographics over a period of 4 days.

BENG ACADEMIC PROJECTS

SunStratos High-Altitude Surveillance Glider, University of Sheffield, UK **Feb 2023 – Jun 2024**
Structures Lead

- Led a **12-member cross-functional team** to design and manufacture a functional prototype; slashed team meeting time by **50%+** through structured task distribution and clearer ownership.
- Oversaw aerodynamic surface design, structural integrity testing, and CAD model development in SolidWorks.
- Trained six recruits to deploy tools effectively, fulfilling deliverables about **33% quicker**.

AeroMAP UAS Build, University of Sheffield, UK **Oct 2023 – May 2024**
Performance Lead

- Directed propulsion and aerodynamic optimisation in a team of 4; reviewed technical drawings and assemblies.
- Applied CFD simulations and FEA to improve aerodynamic surfaces and weight balance for all 5 major components.
- Increased stability by **25%** and flight time by **15%**, enhancing performance during flight testing.

Round-the-Pole Flyer, University of Sheffield, UK **Feb 2023 – Jun 2023**
Team Designer

- Designed and built a high-efficiency single-propeller aircraft prototype, boosting power-to-weight ratio by **35%** using laser-cut plywood and optimised aerofoil geometry.
- Reduced costs by **44%** through efficient manufacturing choices and rapid iteration.

UKSEDS National Rocketry Competition, University of Sheffield, UK **Oct 2022 – May 2023**
Design Team Lead

- Spearheaded rocket design, aerodynamics, and propulsion subsystems for an innovative 2-Whistle Rocket concept.
- Reduced drag coefficient by **22%**; communicated iterative analysis results clearly to drive design decisions.
- Ranked **Top 10 nationally**, outperforming multiple universities.

SKILLS, INTERESTS & EXTRACURRICULAR ACTIVITIES

Simulation & Engineering Tools

ANSYS Suite, OpenFOAM, Tecplot, ParaView, Fidelity Pointwise, SolidWorks, Fusion 360 (FEA & Design), L^AT_EX, Microsoft Office Suite

Programming / Software Development

Python, C++, C#, MATLAB scripting, FORTRAN; Linux environments; automation of analysis and reporting outputs

Manufacturing / Prototyping

3D printing (multi-material), laser cutting, hand fabrication, composites machining

Project Delivery

Gantt timeline control, stakeholder coordination, technical documentation, strong time management and execution under tight timescales

Technical Strengths

Aerodynamics, structural analysis (FEA), heat transfer, system optimisation, data analysis and uncertainty-aware decision support

Professional Memberships

Royal Aeronautical Society (RAeS) Student Affiliate